

1. Administrative & Study Identification

Full Study Title: Phase III Study for Evaluation of the Diagnostic Performance of [18F]CTT1057 PET Imaging in Patients With Prostate Cancer With Rising PSA Levels [Biochemical Recurrence (BCR)] **Short Title/Acronym:** GuidePath **Protocol Number:** CAAA405A12301 **NCT Number:** NCT04838613 **Sponsor Name:** Novartis Pharmaceuticals **Investigational Product:** [18F]CTT1057 (vidoflufolastat (18F)) **Phase of Development:** Phase III **Medical Monitor:** Novartis Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the diagnostic performance of [18F]CTT1057 as a PET imaging agent for the detection and localization of Prostate Specific Membrane Antigen (PSMA) positive tumors in prostate cancer (PCa) patients diagnosed with biochemical recurrence (BCR) after initial definitive therapy. **Secondary Objectives:** To assess the patient-level positive predictive value related to PSA levels, evaluate inter-reader variability and intra-reader reproducibility among independent reviewers, and evaluate the overall safety profile of the radiotracer.

2.2 Background & Rationale Improved diagnostic accuracy in patients with prostate cancer experiencing their first biochemical recurrence (BCR) with low prostate-specific antigen (PSA) levels is critically needed. Conventional imaging techniques often lack the sensitivity required to detect early recurrence. [18F]CTT1057 is a highly selective PSMA-targeted PET radiotracer with a phosphoramidate scaffold. This study aimed to demonstrate whether this next-generation imaging modality can accurately identify local and metastatic recurrence to facilitate more effective clinical interventions.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Single-arm **Blinding:** Open-label (with blinded independent central readers for the PET/CT images) **Randomization:** N/A

3.2 Planned Enrollment & Duration Target \$N\$: Approximately 190 enrolled patients (to ensure at least 152 evaluable for co-primary endpoints) **Number of Sites:** Multicenter (13 locations) **Estimated Study Start:** 30-Sep-2021 **Estimated Primary Completion:** 23-Nov-2023

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Diagnosed with biochemical recurrence (BCR) of prostate cancer after initial definitive therapy (radical prostatectomy [RP] or curative intent radiation therapy [RT]). 2. By AUA criteria for RP: Initial serum PSA of ≥ 0.2 ng/mL measured at least 6 weeks after RP with a second confirmatory persistent PSA level of > 0.2 ng/mL. 3. By ASTRO-Phoenix criteria for RT: Rise of serum PSA measurement of 2 or more ng/mL above the nadir PSA observed post-RT.

4.2 Exclusion Criteria 1. Inability to complete the needed investigational and standard-of-care imaging examinations due to any reason (e.g., severe claustrophobia, inability to lie still). 2. Any additional medical condition indicating a significant risk to safety or impairing study participation (e.g., severe urinary incontinence, New York Heart Association class III or IV congestive heart failure). 3. Prior receipt of prohibited systemic antineoplastic therapy before the completion of PET/CTs and Composite Truth Standard (CTS) procedures.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Patients received a single injection of 370 MBq of [18F]CTT1057. (They also received 150 MBq of [68Ga]Ga-PSMA-11 as part of standard-of-truth imaging). **Route of Administration:** Intravenous injection. **Duration of Treatment:** Single administration followed by PET/CT scan 90 minutes ($\pm$ 30 min) post-injection.

5.2 Concomitant & Prohibited Therapy Permitted: Standard-of-care management not interfering with PSMA imaging. **Prohibited:** Concomitant androgen deprivation therapy (ADT) given before the completion of CTS procedures.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures	:---
:---	:---	Screening	Day -28 to 0 Informed Consent, Medical History, baseline PSA evaluation
Baseline	Day 0		Injection of radiotracers, [18F]CTT1057 PET/CT and [68Ga]Ga-PSMA-11 PET/CT imaging, Safety Observation
Follow-Up	Month 3		Clinical follow-up, PSA measurement, and standard-of-care imaging (CTS Level 2 or 3) to confirm standard of truth

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Region-level correct localization rate (CLR) and patient-level positive predictive value (PPV) of [18F]CTT1057 to detect PSMA-positive lesions.

Measurement Method: Scans assessed by 3 independent blinded readers and compared against a hierarchical Composite Truth Standard (CTS): Level 1 (histopathology), Level 2 (imaging including contrast CT and [68Ga]Ga-PSMA-11 PET/CT), and Level 3 (PSA decline 3 months post-RT).

7.2 Secondary & Exploratory Endpoints Patient-level positive predictive value stratified by PSA levels (e.g., ≤ 0.5 ng/mL, $0.5 - 1.0$ ng/mL, etc.). Inter-reader variability and intra-reader reproducibility of the [18F]CTT1057 PET/CT image assessments.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring during the imaging visit and post-injection follow-up calls to collect injection-site reactions and general AEs. **Serious Adverse Events (SAEs):** Required to be documented and reported per protocol and standard regulatory timelines (typically 24 hours). **Data**

Monitoring Committee (DMC): Overseen by the sponsor to evaluate the safety and tolerability profile of [18F]CTT1057.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Efficacy Evaluable Set (approx. 161 patients) consisting of participants with evaluable [18F]CTT1057 imaging and at least one evaluable CTS assessment. **Sample Size Justification:** A sample of 152 evaluable patients was required so that the lower-bound 95% Confidence Interval (CI) would surpass 50% for region-level CLR and 20% for patient-level PPV for at least 2 of the 3 independent readers. **Handling of Missing Data:** Patients without both an evaluable PET/CT scan and an evaluable CTS assessment were excluded from the primary co-endpoint calculations.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study was conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Site monitoring visits to ensure proper handling of radiopharmaceuticals, adherence to scanning protocols, and accurate data entry. **Audit Trail:** eCRF locking, tracking of CTS hierarchical levels, and data clarification mechanisms managed by a centralized imaging core lab.

11. Study Identifiers & Governance

Primary ID: CAAA405A12301 **Registry Number:** NCT04838613 **Sponsor/Lead Organization:** Novartis Pharmaceuticals **Study Title:** GuidePath **Official Regulatory Title:** Phase III Study for Evaluation of the Diagnostic Performance of [18F]CTT1057 PET Imaging in Patients With Prostate Cancer With Rising PSA Levels [Biochemical Recurrence (BCR)]

12. Clinical Framework (Prostate Cancer / BCR Specific)

Primary Condition: Prostate Cancer Biochemical Recurrence (BCR) **Diagnosis Threshold:** PSA ≥ 0.2 ng/mL post-RP or ≥ 2.0 ng/mL above nadir post-RT **Medical Methodology:** PSMA-Targeted PET/CT Imaging **Optimization Target:** Precise detection and spatial localization of PSMA-positive lesions

13. Study Procedures & Methodology

Management Pathway: Hierarchical Composite Truth Standard (CTS) validation **Education Protocol (HCP):** Site training for imaging acquisition parameters, radiotracer handling, and protocol compliance **Education Protocol (Patient):** Informed consent administration, preparation instructions for PET/CT scan **Quality Audit Mechanism:** Blinded image interpretation by 3 independent central readers to evaluate inter-reader reliability and accuracy

14. Observation & Follow-Up Schedule

Total Duration: 3 months (from baseline to final follow-up imaging/PSA tests) **Onsite Visit Cadence:** Day 0 (Imaging) and Month 3 (Follow-up imaging or clinical assessment) **Remote Monitoring Frequency:** Intermittent monitoring for post-scan adverse events **Checklist Review Interval:** End-of-study CTS validation (Month 3)

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				